

# Linear Algebra PS1

Show equations, substitutions, intermediate reasoning, and a final labeled answer.

1. [8 pts] Solve  $6x+1y=8$  and  $4x+5y=14$ . Show elimination and verify by substitution.

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2. [6 pts] For  $A=\begin{bmatrix} 2 & 2 \\ 2 & 3 \end{bmatrix}$ , compute  $\det(A)$  and state whether  $A$  is invertible. Justify the conclusion.

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3. [8 pts] Let  $T(x,y)=(x+2y, 3x-y)$ . Find  $T(1,-1)$ , write the matrix of  $T$ , and state whether  $T$  is linear.

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4. [6 pts] Give a test for linear independence of vectors and apply it to  $\{(1,0),(0,1)\}$ .

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5. [5 pts] Explain geometrically what the null space of a matrix represents and how it relates to  $Ax=0$ .

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6. [8 pts] Solve  $5x+2y=9$  and  $3x+3y=9$ . Show elimination and verify by substitution.

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